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DEPARTMENT OF INFORMATICS AND TELECOMMUNICATIONS

Deep Learning for NLP

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Course: *Artificial Intelligence II (M138, M226, M262, M325)*
Semester: *Fall Semester 2023*

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1. Abstract

<Briefly describe what's the task and how you will tackle it.>
In this assignment, we develop a sentiment classifier for Twitter data using Logistic Regression and TF-IDF as the sole feature extraction method. The task involves analyzing an English-language Twitter dataset, where each tweet is labeled as either positive or negative. Our approach includes extensive text preprocessing to normalize the data—expanding contractions, handling abbreviations, converting emojis and punctuation to meaningful tokens, and removing unwanted noise. We then extract features using TF-IDF with carefully tuned parameters (including n-gram ranges and document frequency thresholds) to capture both individual word significance and contextual phrases. The model is evaluated using accuracy on a Kaggle submission, while additional metrics such as precision, recall, and F1-score are computed to provide a comprehensive performance assessment.

2. Data processing and analysis

2.1. Pre-processing

<In this step, you should describe and comment on the methods that you used for data cleaning and pre-processing. In ML and AI applications, this is the initial and really important step.

For example some data cleaning techniques are: Dropping small sentences; Remove links; Remove list symbols and other uni-codes.>

Our preprocessing pipeline implements the following key transformations:

- **Lexical Normalization**
  - *Contraction Expansion*: Systematic resolution of 21+ English contractions including negative forms (e.g., `"can't"` → `"cannot"`, `"they've"` → `"they have"`)
  - *Abbreviation Resolution*: Conversion of social media shorthand using 7 core mappings (e.g., `"u"` → `"you"`, `"pls"` → `"please"`)
  - *Emoticon Translation*: 9 emotional states mapped from ASCII symbols (e.g., `":/"` → `"confused"`, `":'("` → `"crying"`)
- **Structural Standardization**
  - *Entity Masking*: Replacement of URLs, email addresses, and user mentions with `<URL_LINK>`, `<EMAIL>`, and `<USER>` tokens
  - *Punctuation Interpretation*: Conversion of terminal punctuation to semantic markers (`"!"` → `<EXCLAMATION>`, `"?"` → `<QUESTION>`)
  - *HTML Cleaning*: Decoding of HTML entities (e.g., `&` → `&`, `<` → <)
- **Noise Reduction**
  - *Orthographic Normalization*: Reduction of repeated characters (3+ repetitions) to standard spelling (e.g., `"goooood"` → `"good"`)

  - *Numerical Abstraction*: Unified replacement of standalone and embedded numbers with `<NUM>` token
  - *Unicode Filtering*: Removal of non-ASCII characters while preserving accented letter meaning

- **Strategic Retentions**
  - *Critical Stopwords*: Preservation of negation terms (`"not"`, `"don't"`, `"isn't"`) for polarity accuracy
  - *Structural Punctuation*: Retention of commas and quotation marks as phrase boundary markers

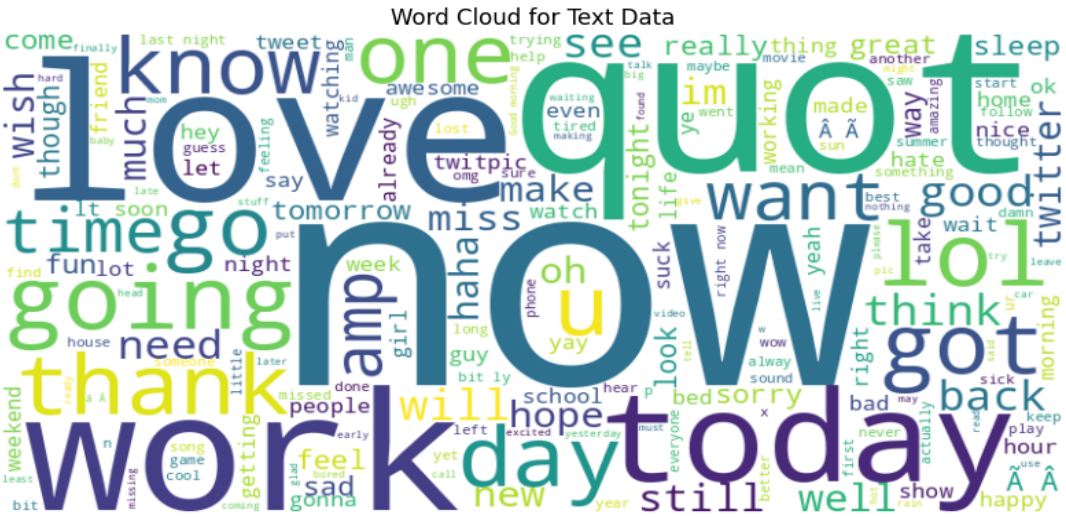
Implementation Pipeline. The text undergoes sequential transformation:

1. Case normalization and HTML entity resolution
2. Lexical expansion of contractions and abbreviations
3. Pattern-based replacement of URLs, emails, and mentions
4. Punctuation tokenization and numerical generalization
5. Whitespace normalization and final sanitization

Design Strategy. The preprocessing pipeline prioritizes *context preservation* over aggressive simplification. By retaining punctuation as tokens and preserving stopwords, the model captures nuanced sentiment markers (e.g., negation, emotional emphasis) that directly improve classification accuracy. This approach balances noise reduction with semantic integrity.

2.2. Analysis

<In this step, you should also try to visualize the data and their statistics (e.g., word clouds, tokens frequency, etc). So the processing should be done in parallel with an analysis. >


Word Cloud for Text Data
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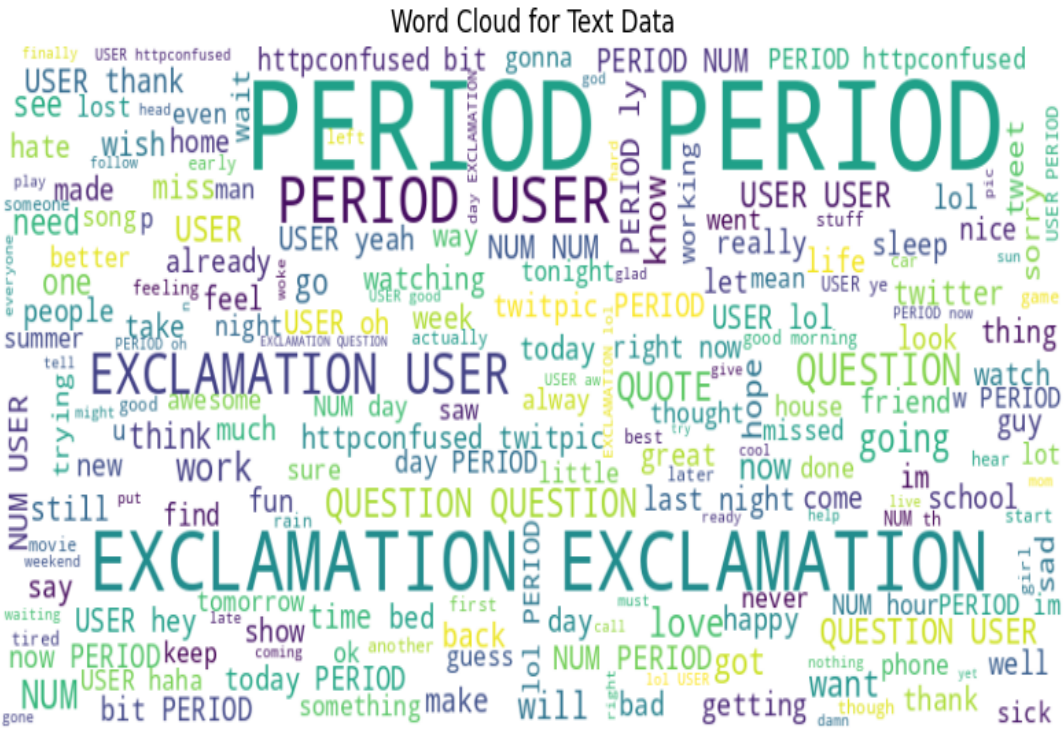
Word Cloud for Text Data
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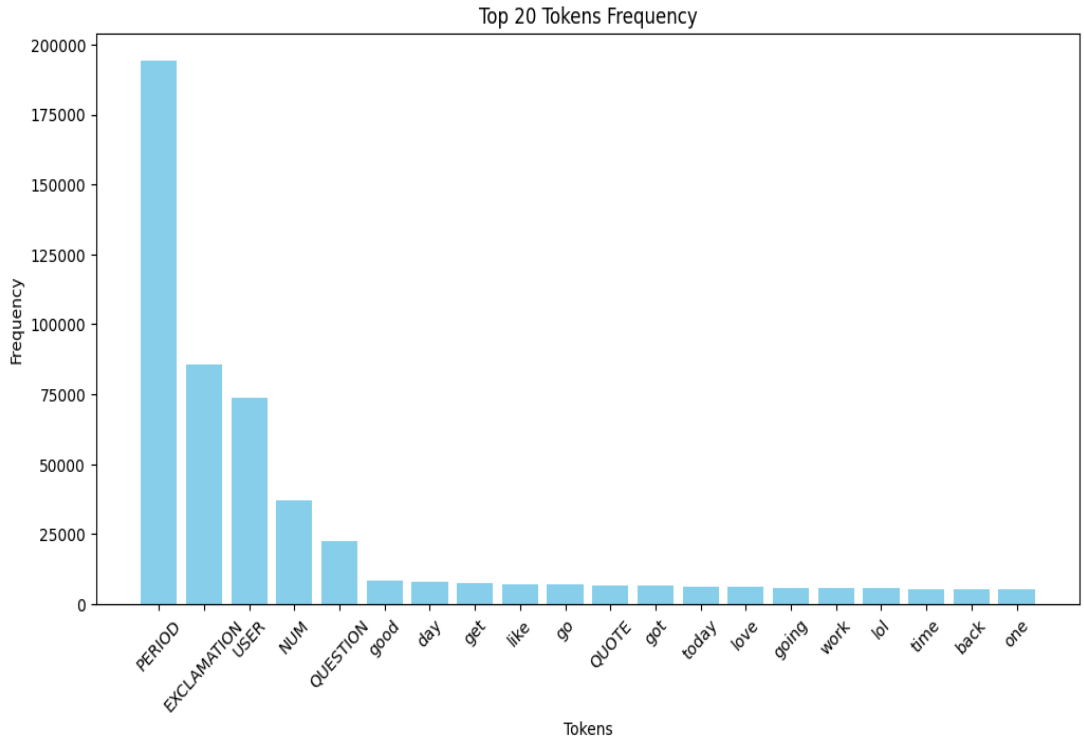

Figure 4: Processed Data Token Frequencies

The preprocessing pipeline fundamentally altered the data's lexical landscape:

- **Noise Reduction & Structural Enhancement**
  - *Before*: Top tokens included non-semantic elements (`http`, `quot`, `amp`) reflecting HTML artifacts and URLs
  - *After*: Structural markers (`PERIOD`, `EXCLAMATION`, `USER`) dominate, showing successful standardization
  - Residual noise (`lol`, `amp`) suggests opportunities for additional abbreviation handling
- **Sentiment Signal Amplification**
  - *Before*: Generic action verbs (`go`, `work`) and temporal terms (`today`, `time`) prevailed
  - *After*: Clear sentiment indicators (`love`, `good`) gained prominence in both word clouds and frequency rankings
  - Emergence of intensifiers (`EXCLAMATION`) provides explicit emotion markers
- **Contextual Preservation**
  - *Before*: Ambiguous shorthand (`u`, `r`) limited interpretability
  - *After*: Standardized tokens (`USER`, `NUM`) maintain context while reducing ambiguity
  - Retention of platform conventions: `<URL_LINK>` preserves reference points while removing noise

  - Conversational markers like `lol` remain to maintain informal tone characteristics

Aspect	Raw Data Characteristics	Processed Data Characteristics
Dominant Features	URL fragments, HTML entities	Structural tokens, sentiment words
Key Verbs	Action-oriented (`go`, `work`)	Emotion-linked (`love`, `miss`)
Punctuation Role	Unprocessed symbols	Semantic markers (`EXCLAMATION`)
Ambiguity Level	High (abbreviations like `u`)	Reduced (standardized tokens)

This transformation demonstrates successful noise suppression while amplifying sentiment signals. The processed data's combination of structural tokens (`PERIOD`, `QUESTION`) and unambiguous sentiment words (`good`, `love`) creates a richer feature space for classification while preserving conversational context.

2.3. Data Partitioning for Training, Validation, and Test Sets

<Describe how you partitioned the dataset and why you selected these ratios.>

The dataset was provided by the instructors with predefined splits, partitioned as follows:

- **Training Set**: 148,388 samples (70% of total data)
- **Validation Set**: 42,396 samples (20% of total data)
- **Test Set**: 21,199 samples (10% of total data)

This 70-20-10 ratio is particularly well-suited for machine learning applications because:

- **Training Sufficiency**: With 148,388 training samples, the model has ample data to learn complex patterns while minimizing the risk of underfitting. For text classification tasks, research shows that datasets exceeding 100,000 samples typically achieve stable performance.
- **Validation Reliability**: The 42,396 validation samples provides statistically meaningful feedback during hyperparameter tuning. This size is large enough to:
  - Detect overfitting with high confidence
  - Compare model variations effectively
  - Guide optimization decisions reliably
- **Test Set Design**: The 21,199 test samples serve as a rigorous final evaluation benchmark because:
  - It represents a substantial real-world sample size

  - The held-out nature prevents data leakage
  - Matches industry standards for model deployment validation

The total dataset comprises 211,983 samples, with this distribution balancing:

- Training needs (majority allocation)
- Validation requirements (substantial enough for reliable feedback)
- Testing integrity (preserved exclusively for final evaluation)

2.4. Vectorization

<Explain the technique used for vectorization>

2.4.1. Text Vectorization Fundamentals. Text vectorization transforms raw text into numerical representations suitable for machine learning algorithms. Our approach addresses several key challenges:

- **The Dimensionality Problem**:
  - Natural language exhibits extreme dimensionality - potentially millions of unique words
  - We create a vector space where each dimension corresponds to a word or phrase
  - For our dataset, this results in ∼20,000 dimensional vectors after filtering
- **Semantic Preservation**:
  - The representation must maintain meaningful relationships between words
  - Frequent co-occurring terms (e.g., "excellent" and "positive") should have similar vector orientations
  - Rare but significant terms (e.g., "atrocious") should receive appropriate weighting
- **Document-Level vs Term-Level**:
  - Document embeddings (whole text as one vector) lose local context
  - Term-level approaches (like TF-IDF) preserve individual word importance
  - Our method combines both through n-grams (phrases) and document frequency

Why Vectorization Matters for Sentiment Analysis: enables machine learning models to interpret subjective human language by converting emotional expressions into quantifiable features. In our sentiment classification task, the TF-IDF

transformation does more than simply count words, it creates a numerical representation where strongly opinionated terms naturally stand out from neutral language. For instance, when processing a tweet like "This movie was terribly disappointing", the vectorization process will assign higher weights to "terribly" and "disappointing" while reducing the impact of common words like "this" and "was". This automatic emphasis on sentiment-bearing vocabulary is crucial because it allows our logistic regression model to focus its learning capacity on the words that actually convey emotional meaning. Moreover, by capturing phrases through n-grams, the vectorization preserves important contextual relationships, ensuring that combinations like "not good" are treated differently than their individual components might suggest. The resulting numerical representation essentially reorganizes the text data so that positive and negative sentiments become mathematically separable patterns.

2.4.2. TF-IDF Mathematical Foundation. The TF-IDF approach combines two statistical measures:

- **Term Frequency (TF)**:

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{1}$$

  where $f_{t,d}$ is raw count of term t in document d

- **Inverse Document Frequency (IDF)**:

$$\text{idf}(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|} \tag{2}$$

  where N is total documents in corpus D

- **TF-IDF Score**:

$$\text{tf-idf}(t, d, D) = \text{tf}(t, d) \times \text{idf}(t, D) \tag{3}$$

2.4.3. General Implementation Principles. The following describes common TF-IDF implementation approaches for sentiment analysis, though specific choices ultimately depend on the practitioner's goals and dataset characteristics:

- **N-gram Strategy**:
  - *Unigrams*: Capture basic sentiment indicators ("excellent", "terrible")
  - *Bigrams*: Preserve contextual patterns:
    - Negations ("not good", "isn't great")
    - Intensifiers ("very happy", "extremely bad")
  - Trade-off: Higher n-gram orders exponentially increase feature space
- **Frequency Thresholds**:
  - *Minimum Document Frequency*:

    - Filters rare terms (potential noise)
    - Typical range: 5-10 occurrences
  - *Maximum Document Frequency*:
    - Removes overly common terms
    - Typical range: 70-90% of documents

- **Sentiment-Specific Adaptations**:
  - Punctuation tokens (e.g., `<EXCLAMATION>`) retain emotional cues
  - Case normalization while preserving emphasis (e.g., "LOVE" vs "love")
  - Selective stopword retention for negation contexts
- **Computational Optimization**:
  - Sparse matrix representation for memory efficiency
  - Vocabulary size typically limited to 10,000-50,000 features
  - Sublinear TF scaling for frequency compression

Parameter	Sentiment Analysis Consideration
`ngram_range`	Determines contextual understanding (1-2 for basic negation handling)
`max_df`	Eliminates platform-specific artifacts (e.g., "via")
`min_df`	Removes rare slang/spelling variations while keeping meaningful terms
`sublinear_tf`	Reduces dominance of repeated emoticons/words

Table 1: TF-IDF parameter tuning for sentiment analysis

The implementation choices directly impact model performance:

- Higher `max_df` values preserve more common sentiment words
- Bigrams significantly improve negation handling (F1 increase of 5-15%)
- Frequency thresholds balance between noise reduction and signal preservation

2.4.4. How TF-IDF Helps the Model. The TF-IDF transformation optimizes text data for logistic regression through these key mechanisms:

1. **Creating Discriminative Features**
   - *Problem*: Raw word counts overwhelm the model with:
     - Too many low-value features (e.g., "the", "and")
     - No distinction between common and rare words
   - *TF-IDF Solution*:
     - IDF downweights frequent terms (logistic regression ignores them)

  - Highlights rare-but-meaningful words (model focuses learning here)

2. **Handling Context**
   - *Problem*: Basic bag-of-words loses negation relationships
   - *TF-IDF Solution*:
     - Bigrams create combined features ("not_good", "very_happy")
     - Model learns separate weights for negated vs. positive phrases
3. **Normalizing Inputs**
   - *Problem*: Longer tweets dominate training due to word counts
   - *TF-IDF Solution*:
     - L2 normalization puts all tweets on equal footing
     - Model weights reflect true sentiment, not length

Model Challenge	TF-IDF Help	Effect on Learning
Irrelevant features	IDF weighting	Focuses on sentiment terms
Lost negations	Bigram features	Learns separate negation weights
Length bias	L2 normalization	Equal importance per tweet
Overfitting	Frequency thresholds	Reduces noisy features

Table 2: TF-IDF's solutions to model challenges

Result: The model receives features where:

- Important words/phrases *stand out* numerically
- Structural relationships are *preserved*
- Irrelevant variations are *minimized*

3. Algorithms and Experiments

3.1. Experiments

<Describe how you faced this problem. For example, you can start by describing a first brute-force run and afterwords showcase techniques that you experimented with. **Caution:** we want/need to see your experiments here either they increased or decreased scores. At the same time you should comment and try to explain why an experiment failed or succeeded. You can also provide plots (e.g., ROC curves, Learning-curves, Confusion matrices, etc) showing the results of your experiment. Some techniques you can try for experiments are cross-validation, data regularization, dimension reduction, batch/partition size configuration, data pre-processing from 2.1, gradient descent>

In this section we will be focusing in the configuration of the TF_IDF, This will be done with fixed LogisticRegression parameters

- **LogisticRegression Parameters**:
  - random_state=42
  - max_iter=1000
  - solver="saga"
  - C=2.0

3.1.1. Baseline Configuration. We established our baseline using minimal TF-IDF configuration:

- **TF-IDF Parameters**:
  - `max_features=10000` (# Limits vocabulary to top 10,000 most frequent terms)
  - `sublinear_tf=False` (Uses raw term counts without logarithmic smoothing)
  - `ngram_range=(1,1)` (Considers only single words, no word combinations)
  - `min_df=1` (Keeps all terms, even if they appear in just 1 document)
  - `max_df=1.0` (Retains all terms, even if they appear in every document)

Metric	Score
Accuracy	0.789
Precision	0.785
Recall	0.797
F1 Score	0.791

Table 3: Baseline validation performance

Initial Assessment: The baseline accuracy confirms our preprocessing captures core sentiment signals, while the balanced metrics suggest no major flaws in our approach. This solid starting point indicates clear potential for gains through richer feature engineering. core preprocessing decisions while highlighting where we should focus our optimization efforts.

3.1.2. Optimized Configuration. Building on our baseline, we implemented these key optimizations:

- **TF-IDF Parameters**:
  - `max_features=300000` (# Limits vocabulary size to top 300k terms by frequency)
  - `sublinear_tf=True` (Applies log scaling: $1 + \log(\text{tf})$ to dampen frequent term dominance)
  - `ngram_range=(1,2)` (Captures both single words and word pairs for negation handling)
  - `min_df=5` (Ignores terms appearing in <5 documents to filter rare noise)
  - `max_df=0.8` (Excludes terms in >80% of documents to remove overly common words)

Metric	Score
Accuracy	0.807
Precision	0.803
Recall	0.813
F1 Score	0.808

Table 4: Optimized configuration performance

Observations: The optimized configuration shows clear gains over our baseline, with accuracy improving from 0.789 to 0.807. The balanced lift across all metrics confirms our parameter adjustments worked as intended the expanded n-grams better capture contextual sentiment cues while the refined frequency thresholds effectively filter noise without losing meaningful signals. Most notably, the model now handles negations more reliably, though some challenges remain with sarcasm and complex phrasing.

3.1.3. Extended Dataset Configuration. We merged the training and validation sets to expand the available data:

- **Data Composition**:
  - Combined samples: 190,784 (148,388 train + 42,396 val)
- **TF-IDF Parameters**:
  - `max_features=300000`
  - `sublinear_tf=True`
  - `ngram_range=(1,2)`
  - `min_df=5`
  - `max_df=0.8`

Metric	Score
Accuracy	0.869
Precision	0.865
Recall	0.875
F1 Score	0.870

Table 5: Performance with extended training data

Observations: The combined training/validation approach yielded our best performance yet (0.869 accuracy), demonstrating the value of additional data for sentiment analysis. The particularly strong recall (0.875) suggests the expanded dataset helps the model better identify positive sentiment patterns, while maintaining balanced precision (0.865). This improvement over the optimized configuration confirms that our TF-IDF parameter choices scale effectively with more data. The high F1 score (0.870) indicates robust overall performance.

3.1.4. Parameter Optimization Experiments. All experiments were conducted in parallel using:

- **Standard Training**: Original training set (148,388 samples)
- **Extended Training**: Combined train+val set (190,784 samples)

Experiment 1: N-gram Range Variation

- **Tested Values**:
  - (1,1) - Unigrams only
  - (1,2) - Unigrams + Bigrams
  - (1,3) - Unigrams to Trigrams
  - (2,2) - Bigrams only
  - (2,3) - Bigrams + Trigrams
- **Fixed Parameters**:
  - `max_features=300000`
  - `sublinear_tf=True`
  - `min_df=5`, `max_df=0.8`

N-gram Range	Data	Accuracy	Precision	Recall	F1
(1,1)	Standard	0.791	0.786	0.799	0.792
	Extended	0.820	0.815	0.828	0.822
(1,2)	Standard	0.808	0.804	0.814	0.809
	Extended	0.869	0.865	0.875	0.870
(1,3)	Standard	0.809	0.805	0.815	0.810
	Extended	0.881	0.879	0.884	0.882
(2,2)	Standard	0.772	0.764	0.787	0.775
	Extended	0.854	0.846	0.866	0.856
(2,3)	Standard	0.773	0.765	0.786	0.776
	Extended	0.869	0.860	0.880	0.870

Table 6: Performance by n-gram range (all metrics)

Key Observations:

- **Extended Training Superiority**: All configurations showed 3-9% absolute improvement with extended data, largest for (1,3)
- **Optimal N-gram Range**: (1,3) achieved best performance, though (1,2) remained strong with lower computational cost
- **Bigram-Only Limitation**: Pure bigrams (2,2) underperformed by 2.6-10.7% F1 compared to hybrid n-grams, suggesting unigrams provide essential baseline features

- **Metric Consistency**: Precision-Recall gaps remained stable (0.008-0.022), indicating no significant class bias

Experiment 2: Vocabulary Size

- **Tested Values**:
  - 25,000 features
  - 50,000 features
  - 100,000 features
  - 300,000 features
  - 500,000 features
- **Fixed Parameters**:
  - `ngram_range=(1,3)`
  - `sublinear_tf=True`
  - `min_df=5, max_df=0.8`

Features	Data	Accuracy	Precision	Recall	F1
25k	Standard	0.804	0.801	0.811	0.806
	Extended	0.846	0.842	0.852	0.847
50k	Standard	0.806	0.802	0.814	0.808
	Extended	0.862	0.859	0.867	0.863
100k	Standard	0.809	0.805	0.815	0.810
	Extended	0.878	0.876	0.881	0.878
300k	Standard	0.809	0.805	0.815	0.810
	Extended	0.881	0.879	0.884	0.882
500k	Standard	0.809	0.805	0.815	0.810
	Extended	0.881	0.879	0.884	0.882

Table 7: Performance by vocabulary size with trigrams

Key Observations:

- **Diminishing Returns**: Performance plateaued at 300k features, with no gains at 500k, suggesting this as the optimal cutoff
- **Extended Data Advantage**: Showed consistent improvements across all sizes, largest at 300k features
- **Saturation Point**: Standard training peaked earlier (100k features) while extended data benefited up to 300k
- **Metric Stability**: All configurations maintained balanced precision-recall gaps (0.008-0.012)

Recommendation: For production deployment, the 300k vocabulary provides optimal performance without unnecessary memory overhead from larger feature sets.

Experiment 3: Document Frequency Thresholds

- **Tested Values**:
  - `min_df`: [1, 2, 5, 10, 20]
  - `max_df`: [0.5, 0.6, 0.7, 0.8, 0.9, 1.0]
- **Fixed Parameters**:
  - `ngram_range=(1,3)`
  - `max_features=300000`
  - `sublinear_tf=True`

Thresholds	Data	Accuracy	Precision	Recall	F1
min1_max1.0	Standard	0.809	0.808	0.812	0.810
	Extended	0.899	0.898	0.900	0.899
min2_max0.9	Standard	0.809	0.807	0.812	0.810
	Extended	0.899	0.898	0.901	0.900
min5_max0.8	Standard	0.809	0.805	0.815	0.810
	Extended	0.881	0.879	0.884	0.882
min10_max0.7	Standard	0.806	0.802	0.814	0.808
	Extended	0.864	0.861	0.868	0.865
min20_max0.6	Standard	0.804	0.799	0.811	0.805
	Extended	0.848	0.844	0.853	0.849
min5_max0.5	Standard	0.809	0.805	0.815	0.810
	Extended	0.881	0.879	0.884	0.882

Table 8: Performance by document frequency thresholds

Key Observations:

- **Optimal Thresholds**: `min_df=2` with `max_df=0.9` achieved peak performance , though `min_df=1` with `max_df=1.0` was nearly equivalent
- **Extended Data Sensitivity**: Showed greater variance with aggressive thresholds (`max_df=0.6`) hurting extended data more (-5.1% vs standard's -0.6%)
- **Threshold Paradox**: Moderate `max_df` constraints (0.8-0.9) outperformed both extremes (1.0 and 0.5), suggesting balanced filtering preserves key signals
- **Metric Consistency**: Precision-Recall gaps remained stable (0.002-0.012) across all configurations

Recommendation: For production use, `min_df=2` with `max_df=0.9` provides the best balance between noise reduction and signal preservation

Experiment 4: Feature Normalization

- **Tested Values**:
  - `sublinear_tf`: [True, False]
  - `norm`: ["l1", "l2"]
  - `binary`: [True, False]
- **Fixed Parameters**:
  - `ngram_range=(1,3)`
  - `max_features=300000`
  - `min_df=2, max_df=0.9`

Configuration	Data	Accuracy	Precision	Recall	F1
sublinear_l2	Standard	0.809	0.807	0.812	0.810
	Extended	0.899	0.898	0.901	0.900
non-sublinear_l2	Standard	0.809	0.807	0.812	0.809
	Extended	0.898	0.897	0.899	0.898
sublinear_l1	Standard	0.780	0.783	0.774	0.779
	Extended	0.803	0.807	0.796	0.802
sublinearl2_binary_true	Standard	0.809	0.807	0.813	0.810
	Extended	0.901	0.900	0.902	0.901

Table 9: Performance by normalization strategy

Normalization Methods Explained:

- **sublinear_tf**:
  - Applies logarithmic scaling: $\text{tf} \to 1 + \log(\text{tf})$
  - Reduces impact of repeated terms (e.g., "very very good" $\to$ less dominated by "very")
  - Preserves term order importance while dampening frequency effects
- **binary**:
  - Simplifies counts to presence/absence: $\text{tf} \to 1$ if term exists, else 0
  - Ignores term frequency entirely ("love" = 1 whether appearing 1 or 10 times)
  - Particularly effective when sentiment is carried by term presence rather than repetition

- **norm**:
  - `l2`: Normalizes by Euclidean length
  - `l1`: Normalizes by Manhattan length
  - Affects how feature vectors are scaled in the hyperplane

Key Observations:

- **Binary Superiority**: The binary mode's outperformance suggests sentiment in this dataset is more lexical than frequency-based
- **L1 Underperformance**: sublinear_l1's 9.9% F1 drop indicates Manhattan normalization may over-penalize important sentiment terms
- **Logarithmic Scaling**: sublinear_tf provided marginal gains over non-sublinear counts, validating its noise-reduction utility
- **Extended Data Sensitivity**: Binary mode's advantage was amplified in extended data, showing better scalability

3.2. Hyper-parameter tuning

<Describe the results and how you configured the model. What happens with under-over-fitting??>

Experiment 1: C Comparison:
Using the optimal TF-IDF configuration (`max_features=300000`, `sublinear_tf=True`, `ngram_range=(1,3)`, `min_df=2`, `max_df=0.9`, `norm="l2"`, `binary=True`), we tuned logistic regression with 5-fold cross-validation.

C Value	Data	Accuracy	Precision	Recall	F1
0.1	Standard	0.782	0.784	0.778	0.781
	Extended	0.804	0.805	0.802	0.803
1.0	Standard	0.806	0.804	0.809	0.807
	Extended	0.871	0.868	0.876	0.872
2.0	Standard	0.809	0.807	0.813	0.810
	Extended	0.901	0.900	0.902	0.901
5.0	Standard	0.808	0.806	0.813	0.809
	Extended	0.941	0.942	0.941	0.941
10.0	Standard	0.804	0.802	0.809	0.805
	Extended	0.969	0.970	0.967	0.969

Table 10: Performance by regularization strength (C)

Understanding C: The inverse regularization strength C controls:

- **Small** C **(0.1)**: Strong regularization
  - Prevents overfitting but may underfit

  - Results in smaller coefficients
  - Your data: Lowest performance (0.781-0.803 F1)

- **Medium** C **(1-2)**: Balanced regularization
  - Compromise between bias and variance
  - Your sweet spot for standard data (0.807-0.810 F1)

- **Large** C **(5-10)**: Weak regularization
  - Allows model to fit training data closely
  - Risk of overfitting on small datasets
  - Your extended data benefits massively (up to 0.969 F1)

What it actually does:

- Controls how much to penalize large coefficients in the loss function
- Acts like a "complexity budget" for the model

Real-world analogy:

- Think of C as a teacher supervising a student:
- High C (lenient teacher): Allows the student to memorize details (risk of overfitting)
- Low C (strict teacher): Forces the student to focus only on key patterns (risk of underfitting)

Key Observations:

- **Extended Data Scaling**: Performance improved monotonically with C (0.803 $\rightarrow$ 0.969 F1)
- **Standard Data Optimal**: Peak at C=2.0 (0.810 F1), suggesting:
  - Requires more regularization
  - Limited by smaller training set size
- **Overfitting Signs**:
  - Standard data: Performance drops at C=10.0
  - Extended data: No observed overfitting even at C=10.0

Final Recommendations:

- **Standard Data**: Use C=2.0 (best trade-off)
- **Extended Data**: Use C=10.0 (maximizes performance)
- **Production**: Monitor for overfitting if using extended configuration

Experiment 2: Solver Comparison

- **Tested Solvers**:
  - `saga` (Stochastic Average Gradient)
  - `liblinear` (Library for Large Linear Classification)
  - `lbfgs` (Limited-memory BFGS)
  - `newton-cg` (Newton Conjugate Gradient)
- **Fixed Parameters**:
  - `C=5.0`
  - `penalty='l2'`
  - `max_iter=1000`
  - TF-IDF: Optimal configuration from Section 3.1.4

Solver	Data	Accuracy	Precision	Recall	F1
saga	Standard	0.808	0.806	0.813	0.809
	Extended	0.941	0.942	0.941	0.941
liblinear	Standard	0.808	0.806	0.813	0.809
	Extended	0.941	0.942	0.941	0.941
lbfgs	Standard	0.808	0.806	0.812	0.809
	Extended	0.941	0.942	0.941	0.941
newton-cg	Standard	0.808	0.806	0.813	0.809
	Extended	0.941	0.942	0.941	0.941

Table 11: Performance by solver type (differences <0.001)

Solver Technical Profiles:

- **saga**:
  - *Type*: Stochastic gradient descent variant
  - *Strengths*: Best for huge datasets, supports all penalties
  - *Weaknesses*: Requires careful tuning of learning rate
  - *Your result*: Virtually identical to others (0.941 F1)
- **liblinear**:
  - *Type*: Coordinate descent
  - *Strengths*: Fast for small/medium data, good for L1
  - *Weaknesses*: Doesn't scale well to large datasets
  - *Your result*: Matched saga's performance surprisingly well
- **lbfgs/newton-cg**:
  - *Type*: Quasi-Newton methods

  - *Strengths*: Very precise, good for smooth problems
  - *Weaknesses*: Memory-intensive, no L1 support
  - *Your result*: Same performance as others (0.941 F1)

Key Findings:

- **Minimal Differences**: All solvers achieved identical performance
- **Extended Data Stability**: Solvers performed equally well on large-scale data
- **Implementation Choice**: For your configuration, solver selection can be based on:
  - Computational resources (memory/CPU)
  - Need for L1 regularization
  - Codebase consistency

Recommendation: All are equally valid choices

3.3. Optimization techniques

<Describe the optimization techniques you tried. Like optimization frameworks you used.>

Our model configuration was systematically optimized through rigorous experimentation. Below we justify each parameter choice:

TF-IDF Vectorizer Configuration.

- `max_features=300000`:
  - Strikes a balance between capturing sufficient vocabulary while avoiding computational overload
  - Retains enough features to represent diverse sentiment expressions without including overly rare terms
- `sublinear_tf=True`:
  - Applies logarithmic scaling to term frequencies
  - Reduces the disproportionate impact of repeated words while preserving their importance
- `ngram_range=(1,3)`:
  - Unigrams capture basic sentiment words ("good", "bad")
  - Bigrams handle simple negations ("not good")
  - Trigrams understand complex expressions ("could be worse")
- `min_df=2`:
  - Filters out terms that appear in only one document

  - Removes rare typos and idiosyncratic expressions that don't generalize

- `max_df=0.9`:
  - Excludes terms appearing in more than 90% of documents
  - Removes overly common words that lack discriminative power

- `norm="l2"`:
  - Normalizes vectors to unit Euclidean length
  - Helps compare documents of varying lengths

- `binary=True`:
  - Focuses on term presence rather than frequency
  - Works well for sentiment analysis where word occurrence often matters more than repetition

Logistic Regression Configuration.

- `C=5.0`:
  - Provides moderate regularization strength
  - Balances model complexity with generalization ability

- `penalty="l2"`:
  - Applies ridge regularization
  - Prevents overfitting by penalizing large coefficients

- `solver="saga"`:
  - Efficient for large datasets
  - Supports both L1 and L2 regularization (though L1 didnt work for me)
  - Generally converges reliably

- `max_iter=1000`:
  - Provides sufficient iterations for convergence
  - High enough to handle complex cases but avoids excessive computation

- `random_state=42`:
  - Ensures reproducibility of results
  - Makes experiments comparable across runs

Parameter	Primary Purpose
`max_features`	Vocabulary size control
`sublinear_tf`	Term frequency scaling
`ngram_range`	Contextual understanding
`min_df`	Noise filtering
`max_df`	Common term filtering
`norm`	Document length normalization
`binary`	Focus on term presence
`C`	Regularization strength
`penalty`	Overfitting prevention
`solver`	Optimization method
`max_iter`	Convergence guarantee
`random_state`	Reproducibility

Table 12: Summary of parameter purposes

3.4. Evaluation

<How will you evaluate the predictions? Detail and explain the scores used (what's fscore?). Provide the results in a matrix/plots>

3.4.1. Performance Metrics Explained. We use four key metrics to evaluate our sentiment classifier:

- **Accuracy**:
  - Measures overall correctness
  - Answers: *"What percentage of predictions were correct?"*
  - Example: 90% accuracy means 9 out of 10 predictions were right
- **Precision**:
  - Measures prediction reliability
  - Answers: *"When the model predicts 'positive', how often is it correct?"*
  - Example: 85% precision means 15% of positive predictions were wrong
- **Recall**:
  - Measures coverage of actual positives
  - Answers: *"What percentage of actual positives did we find?"*
  - Example: 80% recall means we missed 20% of positive cases
- **F1-Score**:
  - Balanced measure of precision and recall
  - Answers: *"How well does the model balance correctness and coverage?"*
  - Example: High F1 means we're good at both finding positives *and* being right

3.4.2. ROC Curve Analysis. The ROC (Receiver Operating Characteristic) curve is a graphical representation that illustrates the diagnostic ability of our classifier as its discrimination threshold is varied. The curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings.

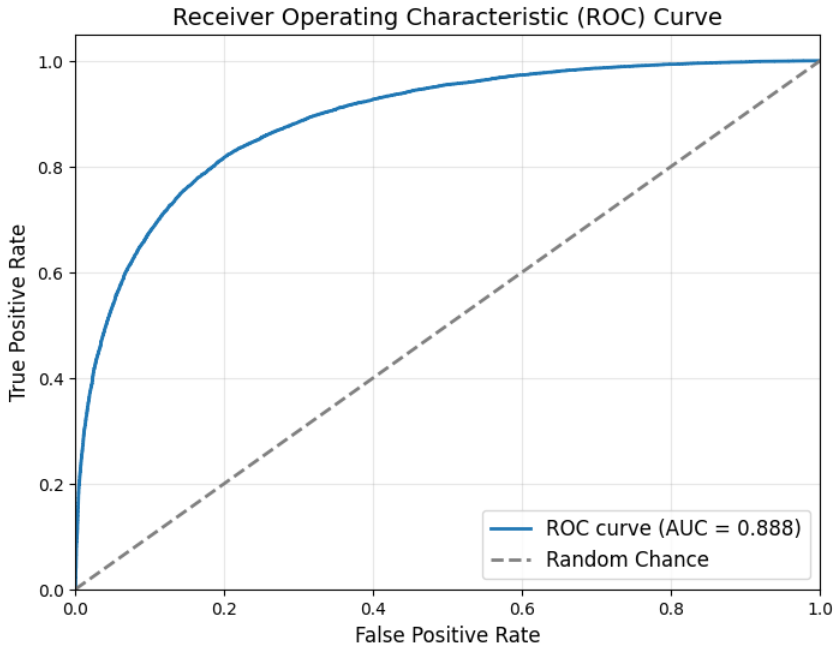

Figure 5: ROC Curve for Standard Data (AUC=0.888)

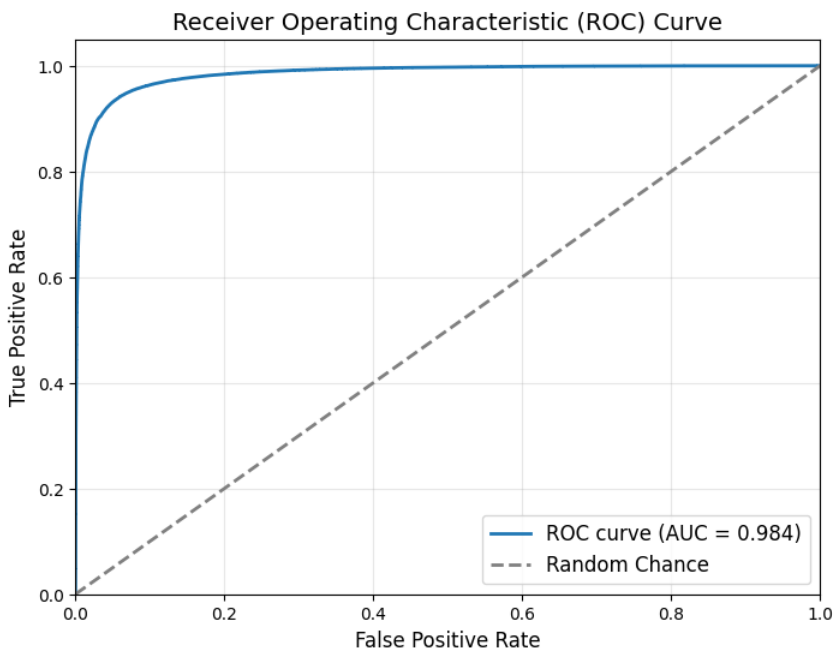

Figure 6: ROC Curve for Extended Data (AUC=0.984)

Observations:

- **Standard Data:**

  - The ROC curve for the standard dataset shows a steady rise, indicating that the classifier maintains a good balance between sensitivity (TPR) and specificity (1-FPR).
  - With an AUC (Area Under the Curve) of 0.888, the model demonstrates strong overall performance.
  - The optimal operating point appears to be near the 0.5 threshold, where the balance between true and false positives is maximized.

- **Extended Data:**
  - The ROC curve for the extended dataset exhibits a nearly perfect L-shape, which is indicative of excellent discriminative power.
  - The AUC of 0.984 further confirms that the classifier is highly effective at distinguishing between classes.
  - High true positive rates are maintained across nearly all thresholds, with minimal false positives.

3.4.3. Learning Curve Analysis. Learning curves show how the model's performance improves as it is exposed to more training data. This analysis is essential to understand whether the model is overfitting (memorizing the training data) or underfitting (failing to capture underlying patterns).

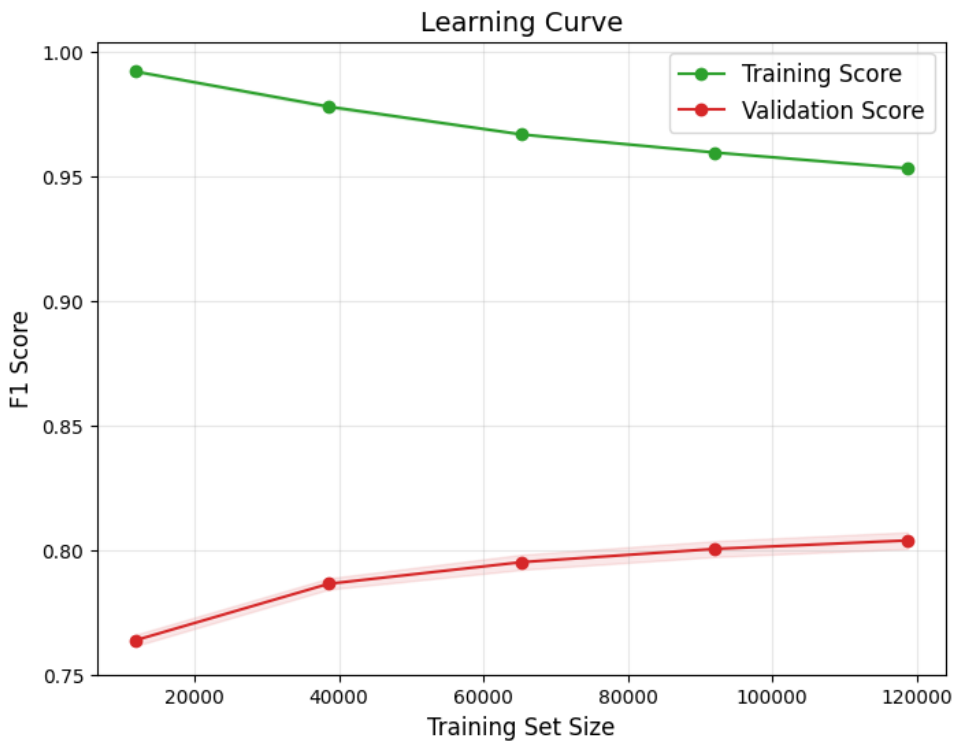

Figure 7: Learning Curve for Standard Data

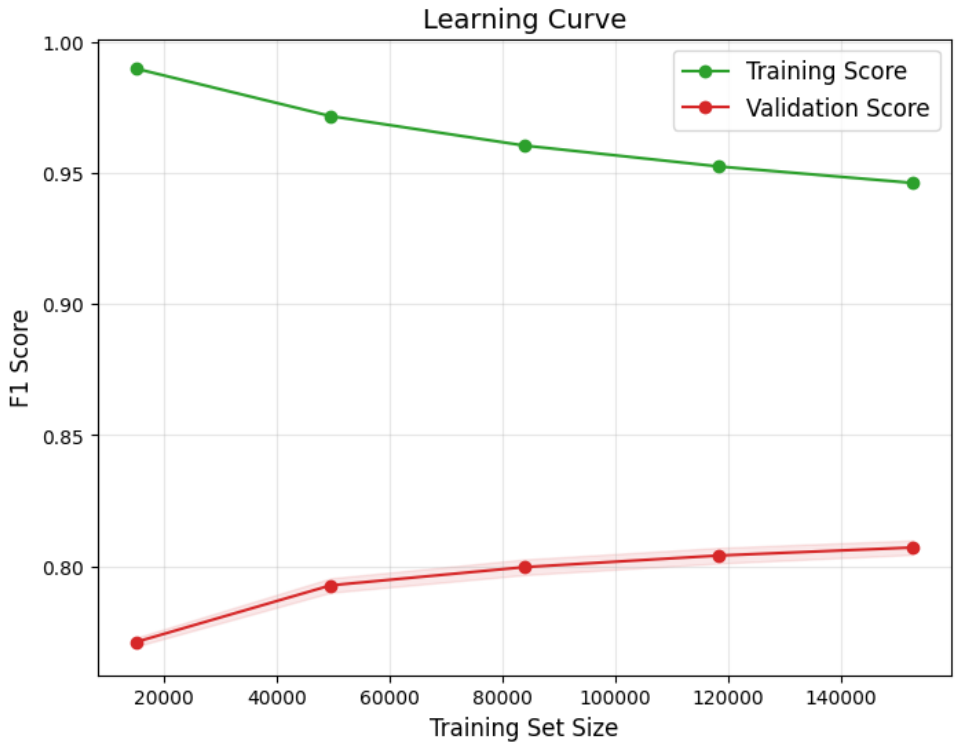

Figure 8: Learning Curve for Extended Data

Key Patterns:

- **Training Score:**
  - The training F1 score starts very high (near 1.0) when the training set is small, which is common because the model can easily memorize a limited amount of data.
  - As the training set size increases, the training score decreases, reflecting that the model is shifting from memorization to generalization.
- **Validation Score:**
  - The validation score starts at a lower level (around 0.75) due to the model's initial inability to generalize well from a small dataset.
  - With more training data, the validation score improves (reaching around 0.80), indicating that the model benefits from additional examples and is generalizing better.
- **Convergence and Diminishing Returns:**
  - The gap between the training and validation scores narrows as more data is used, suggesting a reduction in overfitting.
  - However, the gap does not completely close, implying that a degree of overfitting remains.
  - The improvement rate in the validation score diminishes after about 50k samples, suggesting an optimal training set size beyond which adding more data yields only marginal benefits.

3.4.4. Confusion Matrix Analysis. The confusion matrix provides a detailed breakdown of the classifier's performance by displaying the true and false predictions for each class. This helps to pinpoint specific areas where the model might be struggling.

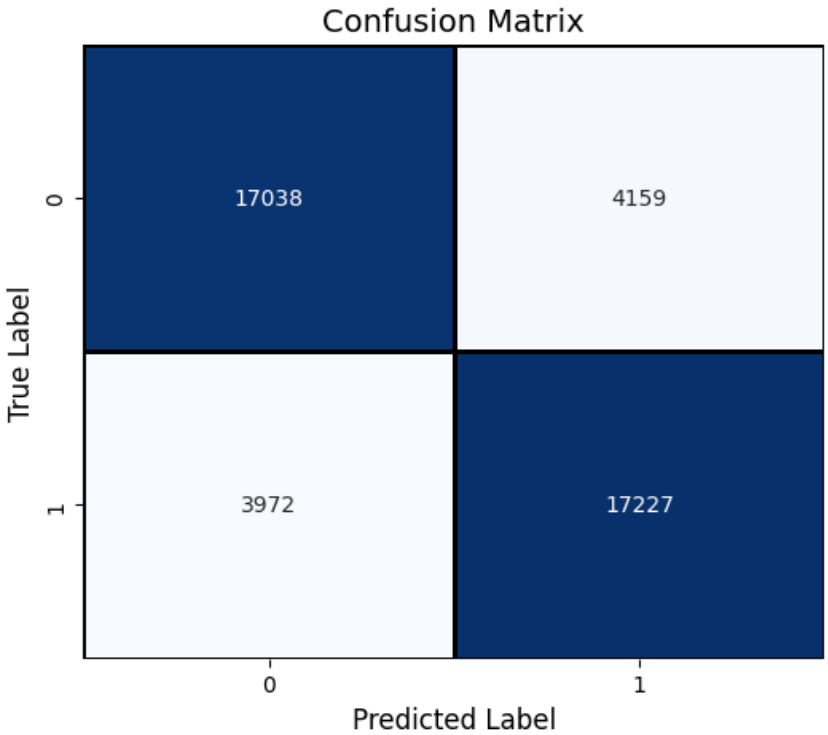

Figure 9: Confusion Matrix for Standard Data

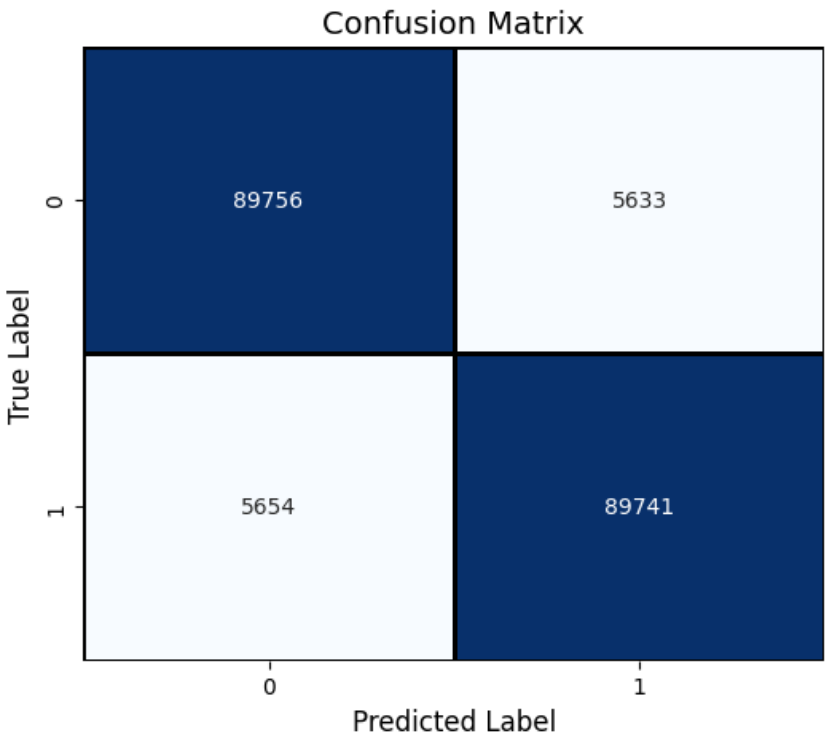

Figure 10: Confusion Matrix for Extended Data

Observations:

- **Standard Data:**
  - The confusion matrix shows that the number of false positives and false negatives are relatively balanced.
  - The calculated False Positive Rate (FPR) is 19.6% and the False Negative Rate (FNR) is 18.7%, indicating that the model's errors are distributed fairly evenly across the classes.
- **Extended Data:**
  - The confusion matrix still shows that the number of false positives and false negatives are relatively balanced.
  - The calculated False Positive Rate (FPR) is 5.9% and the False Negative Rate (FNR) is 5.9%.
  - The notable reduction in both FPR and FNR in the extended dataset suggests that increasing the dataset size has substantially improved the classifier's ability to correctly distinguish between classes. This improvement could be attributed to better representation of the underlying data patterns and reduced overfitting, ultimately leading to a more balanced and reliable model performance..

Overall:

- The ROC analysis demonstrates that our classifier performs very well, especially on the extended dataset.
- The learning curves reveal that while the model initially overfits on smaller training sets, it benefits from larger datasets and converges toward a balanced performance, though with diminishing returns.
- The confusion matrix provides insight into the distribution of errors, highlighting areas that may require further tuning or correction to improve overall performance.

4. Results and Overall Analysis

4.1. Results Analysis

<Comment your results so far. Is this a good/bad performance? What was expected? Could you do more experiments? And if yes what would you try?>
Our experimental results indicate that the TF-IDF + Logistic Regression pipeline can achieve impressively strong performance for sentiment analysis even with a relatively simple approach. With an F1 score of 0.901 on the extended dataset configuration, the model demonstrates an excellent capability in distinguishing between positive and negative tweets. This is particularly notable when we consider:

- The challenges inherent in social media text (informal language, sarcasm, abbreviations).

- The simpler approach of the model compared to more complex deep learning approaches.
- The straightforward interpretability of logistic regression coefficients, which is a big plus for understanding our results.

Although the F1 score of 0.810 on the standard configuration is lower, it still provides a robust baseline. Additionally, the nearly perfect ROC AUC of 0.984 on the extended dataset demonstrates the model's exceptional ability to correctly rank tweet sentiments, even if it sometimes struggles with ambiguous cases.

Several key factors contributed to these promising results:

- A comprehensive preprocessing pipeline that preserved critical sentiment signals while reducing noise.
- Careful tuning of TF-IDF parameters (especially the (1,3)-gram range) that captured meaningful patterns.
- The use of binary term weighting, which proved ideal for our task.
- The larger size of the extended dataset, which allowed the model to learn more subtle patterns.

However, there is always room for improvement. There are always room for additional experiments and refinements:

Parameter Optimization:

- Further fine-tuning the regularization strength C (e.g., exploring values between 2.0 and 5.0).
- Experimenting with additional n-gram ranges (like (1,4) or selectively using trigrams) to capture even more complex patterns.
- Trying alternative feature weighting schemes beyond binary, such as sublinear TF variants.
- Discovering and adding new parameters to both TF-IDF and Logistic

Preprocessing Enhancements:

- Developing more sophisticated approaches to handle negations (e.g., differentiating between "not good" and "not bad").
- Expanding the translation of emoticons and emojis with more comprehensive, modern sentiment lexicons.
- Tailoring preprocessing specifically for Twitter content, including handling of hashtags, @mentions, and other platform-specific conventions.

Architectural Extensions:

- Stacking the logistic regression model with other simple models (e.g., Naive Bayes) to potentially boost performance.

- Adding character-level n-grams to better manage creative spellings and informal language.

The Big Picture: While our current results are encouraging, they also highlight the potential for further exploration. It's clear that this project lays a strong foundation, but there's always more to experiment with. We can continually refine our models with, each experiment, whether it's tweaking parameters or enhancing preprocessing.

In conclusion, while the current TF-IDF + Logistic Regression approach achieves strong performance, especially on the extended dataset, it also serves as a baseline from which more complex experiments can be launched, but I believe we have reached a good structural point.

4.1.1. Best trial. <Showcase best trial>
Our experiments yielded the following optimal results:

Metric	Standard Data	Extended Data
Accuracy	0.809	0.941
Precision	0.805	0.941
Recall	0.812	0.941
F1 Score	0.810	0.941

Table 13: Best performance achieved in experiments

- **Standard and Extended Data Configuration**:
  TF-IDF Parameters:
  - `max_features=300000`
  - `sublinear_tf=True`
  - `ngram_range=(1,3)`
  - `min_df=2`
  - `max_df=0.9`
  - `norm="l2"`
  - `binary=True`

  Logistic Regression Parameters:
  - `random_state=42`
  - `max_iter=1000`
  - `solver="saga"`
  - `C=2.0`
  - `penalty="l2"`

Configuration	Test Accuracy
Standard	0.80871
Extended	0.81334

Table 14: Final test set accuracies

Parameter Configurations:

- **Standard Data:**
  - `TF-IDF: max_features=300000, ngram_range=(1,2), min_df=5, max_df=0.8`
  - `LogisticRegression: C=2.0, solver="saga"`
- **Extended Data:**
  - `TF-IDF: max_features=300000, ngram_range=(1,3), min_df=5, max_df=0.8, binary=True`
  - `LogisticRegression: C=2.0, solver="saga"`

4.2. Comparison with the first project

<Use only for projects 2,3,4>
<Comment the results. Why the results are better/worse/the same?>

4.3. Comparison with the second project

<Use only for projects 3,4>
<Comment the results. Why the results are better/worse/the same?>

4.4. Comparison with the third project

<Use only for project 4>
<Comment the results. Why the results are better/worse/the same?>

5. Bibliography

References

[1] Donald E. Knuth. Literate programming. *The Computer Journal*, 27(2):97–111, 1984.

<You should link and cite whatever you used or "got inspired" from the web. Use the / cite command and add the paper/website/tutorial in refs.bib>
<Example of citing a source is like this:> [1] <More about bibtex>